Manifold-Aware Concept Erasure and Evaluation

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Abstract

Although the ambient representation space of a language model is high-dimensional, the set of sentence representations used in practice can lie on a much lower-dimensional natural manifold. This creates a problem for gradient-based concept erasure: when erasure updates are computed in the full ambient space, they are likely to push representations off that manifold and damage control concepts instead of removing only the target one. We introduce Tangent Gradient-based Concept Erasure (TanGCE), which constrains nonlinear erasure updates to a locally estimated tangent space so that the intervention stays aligned with the geometry actually used by the model. This geometric constraint lets TanGCE safely employ larger erasure strengths λ than ambient updates, improving removal of nonlinear concept information while keeping control-task changes small. Across three text datasets, five language models spanning three families, and two vision benchmarks in CLIP ViT-B/32 representations, TanGCE consistently improves erasure precision over ambient gradient updates while maintaining competitive control-concept preservation. These results support a simple picture: once concept removal is constrained to the natural manifold of representations, stronger interventions become possible without paying the same collateral-damage cost.

1 Introduction

Effective concept erasure in neural representations requires a delicate balance between removing targeted information and preserving the rest of the representation. This is especially difficult for nonlinear concepts: an edit may defeat one trained probe while leaving behind signal that a freshly trained nonlinear classifier can still recover, as emphasized by Elazar and Goldberg's cautionary analysis of post-hoc recovery after adversarial removal [Elazar and Goldberg, 2018]. For erasure to be useful, it must therefore achieve two goals simultaneously: high *erasure precision* on the target concept and high *surgicality* on control concepts.

We argue that a central reason for the utility collapse of ambient-space erasure is geometric. The hidden space of a language model can be high-dimensional, but the image of actual sentence representations need not fill that ambient Euclidean space. Instead, it can concentrate near a much lower-dimensional support. In our primary nonlinear setting, this effective dimension is roughly an order of magnitude smaller than the ambient one. When gradient-based erasure is applied in the full hidden space while the data occupy only this lower-dimensional support, the risk of stepping off the natural representation manifold increases. Once that happens, even a direction that is locally predictive of the target concept can damage unrelated structure simply by pushing the representation into regions the model does not normally occupy.

This observation suggests a different design principle for nonlinear erasure: the edit should follow the local geometry of the representation manifold rather than the full ambient space. Intuitively, if the relevant representation support is modeled locally by a tangent space, then only the tangent component of an erasure update preserves the first-order geometry actually used by the model. Constraining edits to that tangent geometry should make erasure more surgical and, crucially, should allow larger erasure strengths λ before control concepts collapse.

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Our contributions are fourfold:

1. **A geometric formulation of nonlinear concept erasure.** We formalize the mismatch between the full hidden-state dimension and the much smaller effective representation dimension, and argue that off-manifold edits are a key failure mode of ambient gradient-based erasure.
2. **A tangent-constrained nonlinear erasure method.** In Section 4, we introduce a method that projects ambient erasure updates onto locally estimated tangent spaces so that interventions stay aligned with the representation geometry used by the model.
3. **An iterative erasure view motivated by recoverability.** Following Elazar and Goldberg's recoverability insight [Elazar and Goldberg, 2018] and the iterative linear perspective of INLP [Ravfogel et al., 2020], we do not treat success against a single scorer as evidence of complete erasure. Instead, the method is evaluated against freshly retrained nonlinear probes, and the erasure itself is iterated to remove residual recoverable signal.
4. **A supervised evaluation centered on precision and surgicality.** We evaluate on three text datasets and two vision benchmarks using retrained concept probes to measure erasure precision and annotated control labels to measure surgicality.

Empirically, the resulting tangent-constrained method achieves the strongest nonlinear erasure across sycophancy, gender, safety, CelebA visual attributes, and FairFace demographic attributes while maintaining competitive control-concept preservation. Its advantage over the ambient ablation also persists across Qwen2.5-0.5B/1.5B/3B, Gemma-2-2B, and Mistral-7B, indicating that the benefit is not specific to a single small backbone. Section 2 positions the method against prior erasure work, Section 4 gives the method and mechanism hypothesis, Section 5 gives the experimental setup, and Section 6 tests erasure precision and surgicality.

2 Related Work

Concept erasure methods differ mainly in the function class against which they erase and in the geometry they assume. Linear methods such as INLP [Ravfogel et al., 2020] and LEACE [Belrose et al., 2023] target linearly decodable concept information. INLP removes successively exposed linear directions by iterative nullspace projection, while LEACE gives the strongest closed-form linear baseline in our comparisons. These methods are essential reference points, but they do not directly address the case where a fresh nonlinear probe can still recover the concept after linear erasure.

That recoverability concern is central to our framing. Elazar and Goldberg [Elazar and Goldberg, 2018] showed that apparent success against one learned remover or classifier does not imply that the protected information is gone for a newly trained downstream model. Amnesic probing [Elazar et al., 2021] and related intervention-based analyses adopt the same conservative lesson: the meaningful question is what remains recoverable after the intervention, not only what the original scorer no longer captures.

Among nonlinear erasure methods, IGBP [Iskander et al., 2023] is the closest prior baseline because it iteratively trains a nonlinear probe and retracts representations toward the current decision boundary. Our paper differs in two ways. First, AmbGCE is introduced only as the direct ambient-space ablation that keeps the iterative nonlinear scorer loop but removes the manifold constraint. Second, TanGCE is the paper's main method: it projects the erasure update onto a locally estimated tangent space, making the geometric hypothesis itself the main point of comparison.

3 Problem Setup and Geometric Motivation

We first define the representation-editing problem, then introduce the geometric language used throughout the paper. Our notation follows the manifold viewpoint in Tu's introduction to smooth manifolds [Tu, 2011]: we use local coordinate maps, tangent spaces, and differentials only as local approximations, not as theorem-level guarantees about neural representations.

Language model and encoder. Let Σ^* denote the set of all strings over a vocabulary Σ. A language model p defines a distribution over Σ^*. We write $\mathrm{enc}_\ell : \Sigma^* \to \mathbb{R}^d$ for the hidden representation at

layer ℓ, where d is the hidden dimension. For a fixed layer ℓ (omitted when clear from context), the representation of string s is $\mathbf{x} = \mathrm{enc}_\ell(s) \in \mathbb{R}^d$.

Concepts and data. A binary concept is defined by a labeling function $\phi : \Sigma^* \to \mathcal{C}$, where $\mathcal{C} = \{0, 1\}$. Each sample i has representation $\mathbf{x}_i \in \mathbb{R}^d$ and concept label $y_i = \phi(s_i) \in \mathcal{C}$, with an optional control label z_i (e.g., profession for gender erasure). We write $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ for the dataset and $\mathcal{D}_c = \{i : y_i = c\}$ for the class-c subset, with $N_c = |\mathcal{D}_c|$.

The class-conditional representation is the random variable $X_c \sim \mathrm{enc}_\ell(s) \mid \phi(s) = c$, i.e., the distribution of hidden states for strings labeled c.

Concept scorer. A concept scorer $f_\theta : \mathbb{R}^d \to \mathbb{R}$ is a differentiable nonlinear function trained to predict y from $\mathbf{x}$. In the experiments we instantiate it as a two-layer MLP trained with binary cross-entropy, but the geometric construction below depends only on the scorer being differentiable. Its per-sample gradient with respect to the input is $\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i) \in \mathbb{R}^d$, which points in the direction of steepest increase in concept probability at $\mathbf{x}_i$.

Representation manifold. (Yanai:) this seems to be the most important part of this section, but it's somewhat lost in the middle of the section Each layer representation $\mathbf{x} = \mathrm{enc}_\ell(s)$ is formally a point in the ambient hidden space $\mathbb{R}^d$, where d is the full hidden-state dimension. We model the representations that actually occur for real inputs as lying near a smooth embedded submanifold $\mathcal{M} \subset \mathbb{R}^d$, following the local manifold viewpoint of Tu [Tu, 2011]. Its intrinsic dimension is denoted by m, with $m < d$ and typically much smaller than d in our settings, meaning that the local degrees of freedom used by the data are far fewer than the directions available in the ambient hidden space. (Yanai:) why do you discuss modeling here? the section is supposed to describe the setup. this should be separated

Now fix a point $\mathbf{x} \in \mathcal{M}$. Let $U \subset \mathcal{M}$ be a small local neighborhood around $\mathbf{x}$, and let

$$\varphi : U \to \Omega \subset \mathbb{R}^m \tag{1}$$

be a local coordinate map. Here Ω is an open set in the coordinate space $\mathbb{R}^m$. Intuitively, φ assigns m local coordinates to nearby points on the manifold. We choose the coordinates so that $\varphi(\mathbf{x}) = 0$, meaning that the point $\mathbf{x}$ is represented by the origin in local coordinates. The inverse map $\varphi^{-1} : \Omega \to U \subset \mathcal{M} \subset \mathbb{R}^d$ is the corresponding local parametrization back into the ambient hidden space.

The tangent space at $\mathbf{x}$ is the best local linear approximation to the manifold near $\mathbf{x}$. Formally, it is the m-dimensional linear space

$$T_{\mathbf{x}}\mathcal{M} = d(\varphi^{-1})_0(\mathbb{R}^m) \subset \mathbb{R}^d, \tag{2}$$

where $d(\varphi^{-1})_0$ is the differential of the local parametrization φ^{-1} at the origin. In ordinary ML language, this differential can be read as the Jacobian or best local linear approximation of φ^{-1} at 0. It maps an infinitesimal coordinate displacement in $\mathbb{R}^m$ to the corresponding first-order displacement in the ambient space $\mathbb{R}^d$. Because $\mathcal{M}$ is embedded in $\mathbb{R}^d$, we can treat $T_{\mathbf{x}}\mathcal{M}$ as an m-dimensional linear subspace of the ambient hidden space.

Ambient versus tangent directions. For any *ambient vector* $\mathbf{g} \in \mathbb{R}^d$ attached to the point $\mathbf{x}$, the local geometric picture suggests a decomposition

$$\mathbf{g} = \mathbf{g}^\top + \mathbf{g}^\perp, \qquad \mathbf{g}^\top \in T_{\mathbf{x}}\mathcal{M}, \quad \mathbf{g}^\perp \perp T_{\mathbf{x}}\mathcal{M}. \tag{3}$$

Here $\mathbf{g}^\top$ is the tangent component, namely the part of $\mathbf{g}$ that moves along the manifold, and $\mathbf{g}^\perp$ is the normal component, namely the part that points away from the manifold. In this local approximation, the tangent component changes the representation along directions already available in natural data, while the normal component is the first-order source of off-manifold displacement. This is the geometric distinction used later by TanGCE: ambient gradient updates may contain both parts, but tangent-projected updates keep only the component that remains aligned with the local geometry of plausible representations.

Recoverability after erasure. Throughout the paper, concept removal is judged only after training a *fresh* probe on the edited representations. This follows the lesson of Elazar and Goldberg [2018]: erasure succeeds only if a newly trained classifier still fails to recover the concept from $\tilde{\mathbf{x}}$.

Conventions. Index i ranges over samples. $\|\cdot\|$ denotes the Euclidean norm unless stated otherwise.

4 Method: TanGCE

We now introduce *Tangent Gradient-based Concept Erasure* (TanGCE), the method presented in this paper. The key geometric premise is simple: if the representation support is locally approximated by an embedded manifold $\mathcal{M} \subset \mathbb{R}^d$, then the ambient erasure gradient should be edited so that the update moves along the tangent space $T_{\mathbf{x}}\mathcal{M}$ rather than along directions normal to the manifold. Each iteration therefore follows the same four-step pattern: train a fresh concept scorer, estimate a local tangent basis, project the scorer gradient into that tangent space, and apply the resulting constrained update. (Yanai:) present the method in an algorithm figure?

4.1 Local tangent estimation

For each query point $\mathbf{x}_i$, we estimate the tangent space $T_{\mathbf{x}_i}\mathcal{M}$ via local PCA on its k-nearest neighbors. Let $\mathcal{N}_k(\mathbf{x}_i) = \{\mathbf{x}_{j_1}, \ldots, \mathbf{x}_{j_k}\}$ denote the k nearest neighbors of $\mathbf{x}_i$ in the training set. We form secant vectors $\mathbf{s}_{j_m} = \mathbf{x}_{j_m} - \mathbf{x}_i$ and compute the local covariance:

$$\mathbf{C}_i = \frac{1}{k} \sum_{m=1}^{k} \mathbf{s}_{j_m} \mathbf{s}_{j_m}^\top \in \mathbb{R}^{d \times d}. \tag{4}$$

The top r eigenvectors of $\mathbf{C}_i$ form the tangent basis

$$\mathbf{B}_i = [\mathbf{b}_i^{(1)}, \ldots, \mathbf{b}_i^{(r)}] \in \mathbb{R}^{d \times r}, \tag{5}$$

where the intrinsic dimension r is selected automatically by a retained-variance rule. The resulting orthogonal projector onto the estimated tangent space is

$$\mathbf{P}_{T_i} = \mathbf{B}_i \mathbf{B}_i^\top. \tag{6}$$

This is the empirical analogue of the differential-geometric tangent space from Section 3: the neighborhood PCA basis provides a local linear model of the representation manifold around $\mathbf{x}_i$.

4.2 Tangent-constrained update

To define the tangent-constrained update, we start from the ambient unit gradient direction

$$\mathbf{u}_i = \frac{\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i)}{\|\nabla_{\mathbf{x}} f_\theta(\mathbf{x}_i)\|}. \tag{7}$$

TanGCE then defines the tangent-constrained direction by projecting onto the local tangent space:

$$\mathbf{u}_i^{\tan} = \mathbf{P}_{T_i}\, \mathbf{u}_i = \mathbf{B}_i(\mathbf{B}_i^\top \mathbf{u}_i). \tag{8}$$

The projected direction is *not* re-normalized to unit length. Its norm $\|\mathbf{u}_i^{\tan}\| \leq 1$ acts as a confidence weight: when the ambient concept gradient is mostly normal to the manifold, $\|\mathbf{u}_i^{\tan}\| \ll 1$, and the edit is naturally attenuated.

TanGCE then applies the update

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \lambda\, (\mathbf{x}_i^\top \mathbf{u}_i^{\tan})\, \mathbf{u}_i^{\tan}. \tag{9}$$

The effective step size is $\lambda \|\mathbf{u}_i^{\tan}\|^2$, which is maximized when the gradient lies entirely in the tangent space and vanishes when it is entirely off-manifold.

This construction is the core difference between TanGCE and its ambient counterpart. AmbGCE applies the update in the full hidden space, whereas TanGCE restricts the update to tangent directions supported by the local geometry of the representations.

4.3 Why tangent projection permits larger λ

AmbGCE edits $\mathbf{x}_i$ along $\mathbf{u}_i \in \mathbb{R}^d$, which can contain a large normal component relative to $T_{\mathbf{x}_i}\mathcal{M}$. If $\mathbf{u}_i^\perp$ is large, increasing λ pushes the representation away from the local manifold approximation and can distort unrelated structure or preserve recoverable signal. TanGCE removes this normal component at first order by replacing $\mathbf{u}_i$ with $\mathbf{u}_i^{\tan}$. This geometric mechanism can permit operation at larger erasure strengths than ambient updates.

4.4 Iterative TanGCE and relation to baselines

We implement TanGCE using an iterative protocol with probe recycling. Each round begins by training a fresh concept scorer on the current representations, computing per-sample gradients, and applying the tangent-constrained update in Eq. 9. This turns the method into an iterative nonlinear eraser: every new scorer is trained specifically to recover whatever target information survived the previous round. The tangent estimator is also refit in every round to track the evolving local geometry.

The rationale follows Elazar and Goldberg [Elazar and Goldberg, 2018]: success against a single trained classifier is not evidence that the concept has been exhausted. We therefore evaluate every erasure method against a retrained probe after each full erasure procedure, and the erasure itself iterates to remove residual recoverable signal. This nonlinear fresh-scorer loop is the conceptual analogue of iterative linear removal in INLP [Ravfogel et al., 2020], but with a newly trained nonlinear scorer at each round.

This setup also clarifies the relation between TanGCE and the baselines. IGBP [Iskander et al., 2023] is the closest prior nonlinear baseline, performing a one-shot correction to the current decision boundary in each round. AmbGCE is the direct ablation of TanGCE: it keeps the same fresh-scorer loop but applies the update in the full ambient space. TanGCE is the full method, adding the tangent-space restriction.

Two diagnostics verify tangent quality throughout the iterative process. *Tangent Gradient Capture* (TGC) measures the fraction of the leakage gradient lying within the estimated tangent space. *Tangent Neighbor Residual* (TNR) measures the fraction of local secant variation not explained by the tangent basis. High TGC and low TNR indicate that the local linear approximation is sufficiently faithful for the tangent-constrained update to be meaningful.

5 Experimental Setup

We evaluate each erasure method by training all probes from scratch on the intervened representations, so that success cannot be attributed to leakage from the erasure scorer itself.

5.1 Evaluation Protocol

Fresh-probe evaluation. The precision metric is intentionally strict: after every erasure method finishes, we train a *new* probe on the intervened representations. If a freshly trained probe can still recover the concept, then the concept was not actually removed in the sense relevant for downstream use.

Shared validation protocol. Most gradient-based methods share a common erasure strength parameter λ. For AmbGCE and TanGCE, λ scales the ambient or tangent-projected update. Because TanGCE's tangent projection attenuates the effective step size by $\|\mathbf{u}_i^{\tan}\|^2 \leq 1$, a fixed λ across methods would systematically under-erase for the tangent-constrained update. Rather than manually choosing different values, we adopt validation-based hyperparameter selection: all λ-based methods search the same grid $\lambda \in \{0.5, 1.0, 2.0, 4.0, 8.0\}$ and select the λ^* that minimizes validation nonlinear probe accuracy. IGBP has no explicit λ parameter—its step size is determined by each sample's distance to the decision boundary—so it runs until convergence.

Domain-specific controls. For sycophancy we use answer matching as the control concept; for Bias in Bios we use profession; for safety we use helpfulness; and for CelebA and FairFace we use selected low-correlation facial attributes (5 per concept). This lets the evaluation distinguish targeted concept removal from broad representational damage.

Vision benchmarks. For cross-modal evaluation we use two face-image datasets. **CelebA** [Liu et al., 2015] provides 202,599 celebrity images with 40 binary facial attributes; we use the official train/val/test split. **FairFace** [Kärkkäinen and Joo, 2021] provides 108,501 diverse real-world face images annotated with gender (binary), race (7 groups, evaluated as one-vs-rest indicators), and age (Young: $\leq$29 vs. not). We use FairFace's published validation split as our test set, with a random 80/20 train-val split from the provided training images. For both datasets we extract CLIP ViT-B/32

`pooler_output` embeddings (768-dim) and apply all erasure methods to the resulting representation matrix.

5.2 Metrics

Erasure precision. Let h^{NL} be a nonlinear probe and h^{L} a linear probe trained on $\tilde{\mathbf{x}}_i$ to predict the target concept y_i. We report their test accuracies on the intervened representations:

$$\mathrm{Prec}^{\mathrm{NL}}_{\mathrm{erase}} = \mathrm{Acc}(h^{\mathrm{NL}}(\tilde{\mathbf{X}}), y), \qquad \mathrm{Prec}^{\mathrm{L}}_{\mathrm{erase}} = \mathrm{Acc}(h^{\mathrm{L}}(\tilde{\mathbf{X}}), y). \tag{10}$$

Lower values indicate stronger erasure. Because the target concepts in this paper are not assumed to be linearly encoded, the retrained nonlinear probe is the primary precision metric and the linear probe is a secondary diagnostic.

Surgicality on control concepts. Let q be a predictor for an annotated control label c (e.g., profession for gender erasure, helpfulness for safety, or selected low-correlation attributes in CelebA). We report the change in control performance relative to the clean representations:

$$\Delta_{\mathrm{ctrl}} = \mathrm{Acc}(q(\tilde{\mathbf{X}}), c) - \mathrm{Acc}(q(\mathbf{X}), c). \tag{11}$$

Smaller $|\Delta_{\mathrm{ctrl}}|$ means the erasure is more surgical. In CelebA, where each target attribute is paired with multiple controls, we report the mean Δ_{ctrl} across the selected controls.

Together, these metrics ask whether an erasure method removes the target concept while preserving unrelated control concepts.

5.3 Datasets, Models, and Compared Methods

This subsection completes the experimental setup by describing the datasets, models, and comparison methods used to evaluate TanGCE.

5.3.1 Experimental Domains

We evaluate our methods across five blocks:

1. **Sycophancy (Primary Nonlinear Case):** Our central focus is on removing agreement bias in model responses. We use the Anthropic model-written-evals sycophancy suite [Perez et al., 2023], which comprises three subsets: NLP survey, PhilPapers 2020, and political typology quiz. Each example is a multiple-choice question prefaced by a persona bio that asserts a position, with a designated *sycophantic* answer (the one matching the persona's assertion) and one or more *non-sycophantic* alternatives. Because the dataset does not provide pre-assigned per-example concept labels, we assign them by running the model itself: concept $= 1$ if the model's argmax prediction matches the sycophantic answer, concept $= 0$ otherwise. This labelling strategy identifies exactly which examples activate the sycophancy direction in the model being studied, rather than relying on a fixed human annotation. For the surgicality control we use *answer matching*: whether the model selects the non-sycophantic (factually correct) answer. This adapts the standard helpfulness control from safety datasets to the sycophancy setting, where "helpful" behaviour is operationalized as resisting social pressure and answering correctly.
2. **Safety (Secondary Case Study):** We evaluate the surgicality of erasure on content safety using the PKU-SafeRLHF dataset [Ji et al., 2024], testing the preservation of helpfulness alongside safety leakage.
3. **Bias in Bios (Diagnostic Baseline):** We use the Bias in Bios dataset [De-Arteaga et al., 2019] for gender and profession classification as a replication anchor. This setting provides a linear comparison point to contrast against the nonlinear gains of tangent-constrained erasure.
4. **CelebA Visual Attributes (Cross-Modal Generalization):** We apply the same erasure methods to CLIP ViT-B/32 [Radford et al., 2021] pooled embeddings (768 dimensions) of CelebA [Liu et al., 2015] face images. CelebA provides 40 binary facial attributes, enabling a rich surgicality analysis: when erasing one attribute (e.g., *Male*), the remaining 39 serve

as potential controls. We select the 5 least-correlated attributes as controls per concept to ensure that control accuracy changes reflect genuine collateral damage rather than attribute correlation.

5. **FairFace Demographic Attributes (Cross-Dataset Vision Replication):** We additionally evaluate on FairFace [Kärkkäinen and Joo, 2021], using CLIP ViT-B/32 pooled embeddings for gender, race, and age concepts. This benchmark tests whether the vision-domain pattern observed on CelebA survives on a more demographically balanced dataset rather than a celebrity-image slice.

5.3.2 Models

Language models. The main seven-method comparison is run on Qwen2.5-0.5B [Qwen Team, 2024] (24 decoder layers, hidden dimension 896) at layer 8. To test whether the tangent-vs.-ambient gap persists across families and scales, we additionally compare TanGCE and AmbGCE on Qwen2.5-1.5B, Qwen2.5-3B, Gemma-2-2B, and Mistral-7B-v0.1, again at layer 8.

Vision encoder. The CelebA and FairFace experiments use CLIP ViT-B/32 pooled embeddings (768 dimensions), extracted once and cached. We keep the encoder fixed across the two vision benchmarks so that the comparison isolates dataset effects rather than encoder differences. (Yanai:) same here: more models

5.3.3 Compared Methods

Methods. The core quantitative comparison evaluates TanGCE against six baselines and ablations: two random baselines (global and per-sample direction), linear projection, INLP [Ravfogel et al., 2020], LEACE [Belrose et al., 2023], and AmbGCE. IGBP [Iskander et al., 2023] is the closest prior nonlinear baseline and serves as the main methodological reference point for the GCE family: it iteratively trains two-class MLP probes and applies Newton-step boundary corrections. AmbGCE is our direct ablation of TanGCE: it uses the same scorer-training loop but applies the update in the full ambient space. TanGCE adds the key ingredient of the paper by estimating local tangent spaces with PCA-based neighborhoods and restricting the erasure direction to a local on-manifold approximation rather than the full ambient space. (Yanai:) add the baselines we discussed on slack

Comparison method definitions. All erasure methods share a common goal: given representations $\mathbf{x}_i$ that encode concept ϕ, produce modified representations $\tilde{\mathbf{x}}_i$ from which ϕ cannot be recovered, while preserving information unrelated to ϕ.

Linear projection. Linear projection removes a single concept direction, typically estimated from the class-conditional means. It is the simplest global linear baseline and cannot remove nonlinear residual structure.

INLP (Iterative Nullspace Projection). INLP [Ravfogel et al., 2020] extends linear projection by repeatedly training linear classifiers on the residual and removing newly exposed linear directions. It is the canonical iterative linear baseline.

LEACE (Least-squares Concept Erasure). LEACE [Belrose et al., 2023] is the strongest closed-form linear baseline in our comparisons. It performs a minimal linear intervention while eliminating linearly decodable concept information.

IGBP (Iterative Gradient-Based Projection). IGBP [Iskander et al., 2023] is the closest prior nonlinear baseline. It iteratively trains a nonlinear probe and moves each representation toward the current decision boundary by a Newton-style correction.

AmbGCE. AmbGCE is the direct ambient-space ablation of TanGCE. It keeps the same nonlinear scorer loop but applies the update in the full hidden space rather than projecting it onto a local tangent approximation.

Table 1: Comparison of concept erasure methods by family and role in this work.

Method	Family	Role in Paper
Linear Projection	Linear	Global baseline
INLP	Linear	Iterative linear baseline
LEACE	Linear	Strong matched linear baseline
IGBP	Nonlinear	Closest prior nonlinear baseline
AmbGCE	Nonlinear	Ambient ablation of TanGCE
TanGCE	Nonlinear	Main contribution

Table 2: Full erasure method comparison at layer 8 across three concept domains. Each λ-based method's λ^* is selected from the same grid $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy, ensuring a fair comparison where every method gets its best shot at erasure; non-λ baselines use their standard procedure. L/NL Leak: retrained linear logistic-regression / nonlinear MLP probe accuracy on the target concept (lower = stronger erasure precision). ΔCtrl: downstream control-task accuracy change relative to clean representations (closer to zero = more surgical). Model: Qwen2.5-0.5B. Control tasks: answer matching (sycophancy), profession classification (gender), helpfulness (safety).

	Sycophancy			Gender (Bias in Bios)			Safety		
Method	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	.685	.748	.000	.719	.758	.000	.798	.805	+.003
Random Per-Samp.	.621	.745	.000	.723	.743	+.003	.784	.799	$-.010$
Linear Proj.	.683	.740	.000	.722	.747	+.001	.807	.800	$-.003$
INLP	.683	.734	.000	.722	.737	+.001	.807	.795	$-.004$
LEACE	.681	.681	.000	.719	.739	$-.002$	.801	.785	$-.015$
AmbGCE	.714	.703	.000	.663	.701	.000	.780	.778	$-.040$
TanGCE	**.628**	**.602**	$-.015$	**.603**	**.618**	$-.008$	**.654**	**.663**	+.039

6 Results

We evaluate seven erasure methods across three concept domains—sycophancy, gender (Bias in Bios), and safety—at layer 8 of Qwen2.5-0.5B. Each method's erasure strength λ^* is selected from the same shared validation protocol described in Section 5. Our main metrics are erasure precision, measured by retrained probe accuracy on the target concept, and surgicality, measured by change in control-task performance.

6.1 TanGCE Achieves the Best Erasure Precision Across All Domains

Table 2 shows that TanGCE achieves the lowest nonlinear leakage across all three domains: 0.602 on sycophancy (vs. 0.681 for LEACE, 0.703 for AmbGCE), 0.618 on gender (vs. 0.701 for AmbGCE), and 0.663 on safety (vs. 0.778 for AmbGCE). Linear methods (Linear Projection, INLP) barely reduce leakage below random baselines, confirming that concept information is predominantly encoded nonlinearly.

6.2 TanGCE Remains Surgical on Control Concepts

Table 2 reports ΔCtrl, the change in downstream control task accuracy relative to clean representations. For sycophancy (answer matching), TanGCE incurs a small -1.5pp drop; all other methods preserve perfect accuracy. For gender (profession classification), TanGCE drops only -0.8pp (0.531 vs. 0.539 clean). On safety, TanGCE *improves* helpfulness ($+3.9$pp), suggesting that removing safety-concept directions frees capacity for the helpfulness task.

6.3 The Lambda Sweep: On-Manifold Updates Support Larger Interventions

(Yanai:) this sounds like an analysis (the part about lambda)

Table 3 isolates the effect of step size by running only AmbGCE and TanGCE at three fixed λ values $(0.5, 4.0, 8.0)$ on the same three domains as above (Qwen2.5-0.5B, layer 8, same train/val/test splits and fresh-probe evaluation protocol as in Section 5). As λ increases, AmbGCE becomes unstable: its

Table 3: λ sweep comparing AmbGCE and TanGCE at matched step sizes across three concept domains (Qwen2.5-0.5B, layer 8). NL: retrained nonlinear probe accuracy on the target concept (lower = stronger erasure). AmbGCE deteriorates at high λ (rising leakage), while TanGCE continues improving, supporting the claim that on-manifold updates can safely employ larger erasure strengths. Bold: best NL per domain.

Concept	Method	NL↓ at $\lambda =$ 0.5	4.0	8.0
Sycophancy	AmbGCE	.678	.694	.734
	TanGCE	.662	.596	**.588**
Gender	AmbGCE	.666	.700	.702
	TanGCE	.704	.580	**.470**
Safety	AmbGCE	.664	.776	.770
	TanGCE	.748	.680	**.646**

Table 4: CLIP ViT-B/32 concept erasure on CelebA (validation HPO for λ). Each method's λ^* is selected from the same grid $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy, ensuring a fair comparison. L/NL Leak: retrained linear / nonlinear probe accuracy on test (lower = better). ΔCtrl: mean control-task accuracy change across 5 least-correlated attributes (closer to zero = more surgical). Representation: CLIP ViT-B/32 pooled embedding (768 dimensions).

Method	λ^*	Male L↓	Male NL↓	Male ΔCtrl	Smiling L↓	Smiling NL↓	Smiling ΔCtrl	Blond_Hair L↓	Blond_Hair NL↓	Blond_Hair ΔCtrl
Random Global	0.5	.997	.990	-0.3	.903	.903	-0.3	.937	.923	-0.3
Random Per-Samp.	0.5	.993	.990	-0.1	.867	.887	-1.3	.933	.920	$+0.1$
Linear Proj.	1.0	.987	.987	-0.7	.903	.913	$+0.1$	.923	.940	-1.4
INLP	1.0	.587	.987	-0.2	.580	.740	$+0.6$	.713	.917	$+0.7$
LEACE	0.5	.997	.997	-0.3	.503	.820	$+0.4$	.903	.917	-0.1
AmbGCE	0.5	.997	.997	-0.9	.787	.877	$+0.1$	.910	.923	-1.1
TanGCE	**8.0**	**.800**	**.847**	-1.1	**.487**	**.443**	-0.1	**.693**	**.720**	-1.6

nonlinear leakage worsens sharply at high step sizes, rising to 0.734 on sycophancy, 0.702 on gender, and 0.770 on safety at λ=8. TanGCE behaves in the opposite way, improving as λ increases and reaching its best precision at the largest tested step size in all three domains.

This pattern is consistent with the manifold hypothesis motivating the method. If the ambient gradient has a substantial off-manifold component, then increasing λ does not simply erase more target information; it also pushes the representation into regions where unrelated structure is distorted. Projecting onto the local tangent approximation suppresses those off-manifold components, which is why TanGCE can safely use larger erasure strengths.

6.4 Cross-Modal Generalization: CelebA Visual Attributes

To test whether TanGCE generalizes beyond language model representations, we apply all seven erasure methods to CLIP ViT-B/32 `pooler_output` embeddings (768-dim) of CelebA face images (see Section 5 for preprocessing and split details). CelebA provides 40 binary facial attributes, giving a rich surgicality profile: when erasing one attribute, the remaining attributes serve as controls. The same validation-HPO protocol ($\lambda^* \in \{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL accuracy) is applied uniformly across all methods.

We treat this cross-modal block as a generality check rather than the primary evidence block for the paper's central claim.

Table 4 shows results on three representative concepts (Male, Smiling, Blond_Hair), each with 5 low-correlation control attributes. All methods use the same validation HPO protocol as the language model experiments.

Table 5 summarizes results averaged over all 40 CelebA attributes. The median selected λ^* column confirms that TanGCE consistently selects higher λ values than other methods—enabled by the

Table 5: CelebA all-40-attribute summary (CLIP ViT-B/32, validation HPO for λ). Each cell averages over all 40 binary CelebA attributes. λ^* is selected per method per concept from the grid $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL probe accuracy. NL Leak: mean nonlinear test accuracy across 40 concepts (lower = better). Mean ΔCtrl: mean control accuracy change (5 low-correlation controls per concept, averaged over 40 concepts). Worst ΔCtrl: worst single control drop averaged over 40 concepts.

Method	NL Leak↓	Mean ΔCtrl	Worst ΔCtrl	Med. λ^*
Random Global	0.883	-0.4%	-1.6%	2.0
Random Per-Samp.	0.882	-0.3%	-1.2%	0.5
Linear Proj.	0.869	-0.8%	-2.7%	4.0
INLP	0.859	-3.5%	-6.9%	1.0
LEACE	0.839	-0.2%	-1.2%	2.0
AmbGCE	0.875	-0.6%	-2.8%	0.5
TanGCE	**0.707**	-0.7%	-2.1%	**8.0**

tangent projection's attenuation of off-manifold displacement—while maintaining competitive control accuracy. Per-concept breakdowns for all 40 attributes (nonlinear leakage and mean ΔCtrl per method) are reported in Appendix A.4, Tables 12 and 13.

6.5 Geometric Characterization: Low-Rank Subspace Isolation and Manifold Curvature

The preceding results establish that TanGCE achieves stronger nonlinear erasure than AmbGCE. This subsection provides the geometric evidence behind that advantage, organized around two distinct observations that operate at different levels of the geometry. The first is a *global linear* property: each domain's representations concentrate near a low-dimensional PCA subspace, and sycophancy's subspace is geometrically isolated from those of other tasks. The second is a *local nonlinear* property: sycophancy's subspace is curved, meaning the locally estimated tangent direction at each sample differs substantially from the globally fitted PCA direction. The first observation explains why targeted erasure is feasible and why ambient-space edits risk spurious interference with unrelated tasks. The second explains why global linear methods (LEACE, Linear Projection) are insufficient for sycophancy and why a locally-estimated tangent constraint is specifically needed.

Experimental setup. We run two diagnostic experiments on layer 8 representations of Qwen2.5-0.5B across five domains: Bias in Bios (gender), code generation, mathematical reasoning, content safety, and sycophancy. For each domain we collect 500 labeled representations and estimate the global intrinsic dimension m via a retained-variance rule applied to a held-out reference set of 2,000 samples (not used in erasure training). All geometry experiments use the same evaluation split as the main comparison.

Q1: Cross-domain subspace isolation. For each ordered pair of domains $(\mathrm{src}, \mathrm{tgt})$, we fit PCA on *target*-domain representations to dimension $k \approx m_{\mathrm{tgt}}$, project *source*-domain representations onto the resulting linear subspace, and measure top-1 next-token agreement and KL divergence relative to the unmodified source representations. High agreement indicates that source representations lie within the target's principal subspace; collapse indicates near-orthogonality of the two subspaces.

Figure 1 (left) shows the 5×5 transfer matrix. The diagonal is uniformly high (top-1 ≥ 0.88), confirming that each domain's principal subspace captures the directions that matter for its own predictions. Off the diagonal, sycophancy is the structural outlier: projecting sycophancy representations onto the biasbios subspace yields top-1 agreement of 0.002 ($\mathrm{KL} = 7.4$), and onto the code subspace gives 0.13 ($\mathrm{KL} = 6.6$). No other pair approaches this degree of isolation. Conversely, math and safety share substantial subspace overlap (mutual top-1 ≈ 0.88–0.92), consistent with those tasks drawing on shared general-language structure. The pattern is stable across layers 4, 8, and 16 (Appendix A).

This isolation has a direct implication for ambient-space erasure. An edit applied in the full hidden space will have nonzero projections onto the biasbios and code subspaces—directions that carry essentially no sycophancy signal. TanGCE attenuates those components by restricting the update to

Table 6: Geometry contrast between sycophancy and gender across completed layers 4, 8, and 12. This is a focused contrast between the primary nonlinear case and the diagnostic baseline, not an all-dataset geometry census. Lower intrinsic dimension and larger local-vs-global angle indicate stronger nonlinearity: a concept whose erasure direction varies significantly across the input space, making a single fixed direction insufficient to capture the full concept. Sycophancy shows substantially higher nonlinearity (dim 34–53, angle 76.9–82.6°) compared to gender (dim 365–461, angle 51.8–55.8°), supporting sycophancy as the primary case for manifold-aware erasure methods. This table demonstrates the geometric contrast but does not establish causality: it does not prove that the observed nonlinearity will be converted into lower displacement or higher utility preservation by any particular erasure method.

Concept	Local intrinsic dimension	Mean local/global angle
Sycophancy	34–53	76.9–82.6°
Gender	365–461	51.8–55.8°

the locally estimated subspace; AmbGCE applies the full ambient gradient, including its projections onto these geometrically irrelevant directions.

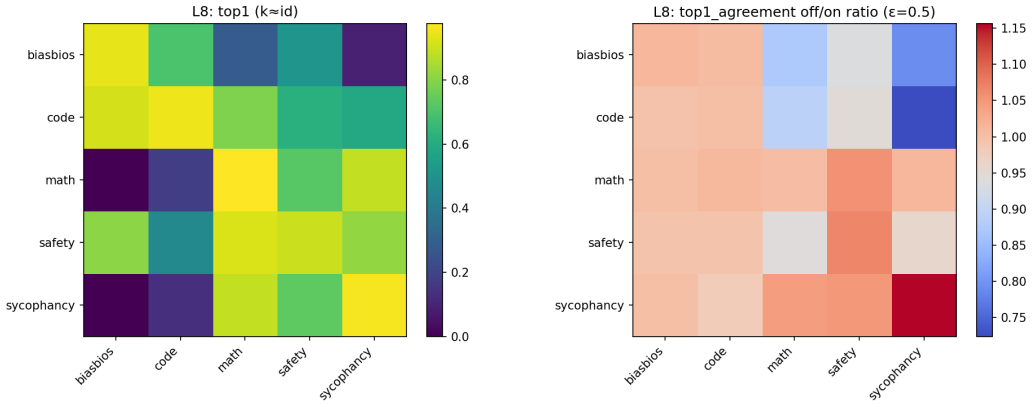
Q2: Subspace sensitivity concentration. To verify that the identified PCA subspaces are causally load-bearing and not merely high-variance summaries, we perturb each sample by $\epsilon\|\mathbf{x}\|$ in three directions: a random direction within the principal subspace (on-subspace), a random direction orthogonal to it (off-subspace), and a random ambient direction. A ratio $\text{off-top1}/\text{on-top1} > 1$ indicates that on-subspace perturbations are more disruptive than equal off-subspace perturbations—the subspace concentrates task-relevant computation.

Figure 1 (right) shows the off/on ratio at $\epsilon = 0.5$ for all 25 domain pairs. Along the diagonal, all five ratios exceed 1, confirming subspace causal relevance across domains. The effect is sharpest for sycophancy (ratio $= 1.16$): an on-subspace perturbation is 16% more destructive than an equal off-subspace one, the largest asymmetry among the five domains. The off-diagonal cells in the sycophancy column (rows 1–2) show ratios of 0.79 and 0.72: the sycophancy subspace is a null direction for biasbios and code predictions, so perturbations within it leave those tasks unaffected while perturbations orthogonal to it do not. Ambient erasure updates that project onto the sycophancy subspace therefore interfere with the orthogonal structure those tasks actually use.

Manifold curvature: why global subspaces are insufficient. Q1 and Q2 establish that representations concentrate in a causally relevant low-dimensional linear subspace. A natural response would be to simply remove the global PCA direction of sycophancy—the approach taken by LEACE and Linear Projection. But this strategy implicitly assumes that the subspace is *flat*: that the first global PCA direction provides a reliable approximation to the locally relevant concept direction at any given sample. This assumption can be tested.

Table 6 measures *manifold curvature* via the local-vs-global angle: the angle between the locally estimated tangent direction at each sample (estimated by neighborhood PCA, exactly as in TanGCE) and the globally fitted first principal component. For a flat linear subspace this angle is $0°$. For a curved manifold it grows as the locally relevant direction rotates across the input space. Sycophancy shows local-vs-global angles of 76.9–$82.6°$ across layers 4, 8, and 12—nearly orthogonal. Gender shows 51.8–$55.8°$, indicating a more planar structure. The quantitative implication is immediate: a method that fits a single global concept direction and applies it uniformly is using a direction that is, on average, $\sim 80°$ away from the direction that actually encodes the sycophancy concept at any individual sample. TanGCE's neighborhood PCA re-estimates this direction locally, recovering the component that globally-fitted methods miss entirely.

This explains the asymmetry in Table 2: LEACE achieves better surgicality on sycophancy because it removes a smaller volume of representation space (one global direction), while TanGCE achieves lower nonlinear leakage because its locally varying direction tracks where the concept actually encodes, even as that direction rotates across the input. The two methods target different levels of the same underlying geometry: LEACE removes the flat global projection; TanGCE follows the curved local manifold.



(a) Q1: cross-domain linear subspace transfer (top-1 next-token agreement, $k \approx m_{\text{tgt}}$). Each cell measures how well source-domain representations are preserved when projected onto the global PCA subspace of the target domain. High diagonal values confirm per-domain subspace self-sufficiency. Sycophancy is the structural outlier: near-zero off-diagonal transfer to biasbios (0.002) and code (0.13) indicates that sycophancy's principal subspace is nearly orthogonal to those of other tasks.

(b) Q2: subspace sensitivity (off/on top-1 agreement ratio, $\epsilon = 0.5$). Values above 1 (red) indicate that on-subspace perturbations are more disruptive than equally sized off-subspace perturbations, confirming the subspace as the primary carrier of task-relevant computation. Sycophancy diagonal (1.16, darkest red) shows the strongest sensitivity asymmetry. Blue off-diagonal entries in the sycophancy column (0.79, 0.72) confirm that the sycophancy subspace is a null direction for biasbios and code predictions.

Figure 1: Geometric diagnostics at layer 8 of Qwen2.5-0.5B across five domains (biasbios, code, math, safety, sycophancy). Rows index the source domain whose representations are used; columns index the domain whose global PCA subspace is applied. **Left (Q1):** linear subspace transfer matrix. **Right (Q2):** subspace sensitivity ratio. Both experiments use 500 evaluation samples per domain with 2,000 additional held-out samples for intrinsic-dimension estimation.

Summary. The two observations connect to different aspects of the TanGCE design. Subspace isolation (Q1/Q2) motivates restricting edits to the directions where sycophancy information actually lives rather than editing the full ambient space. Manifold curvature (Table 6) motivates estimating those directions locally at each sample rather than fitting a single global direction. Together they form the geometric case for a method that is both subspace-restricted and locally adaptive—precisely the design of TanGCE.

6.6 Multi-Model Generalization

To assess whether TanGCE's advantage is specific to Qwen2.5-0.5B or holds across model families and scales, we repeat the evaluation at layer 8 across five models spanning three families and parameter counts from 0.5B to 7B: Qwen2.5 (0.5B, 1.5B, 3B), Gemma-2-2B, and Mistral-7B-v0.1. We compare only TanGCE and AmbGCE, the two methods that differ solely in their gradient constraint, isolating the effect of tangent-space projection. The same validation HPO protocol ($\lambda^* \in \{0.5, 1.0, 2.0, 4.0, 8.0\}$, selected by lowest validation nonlinear probe accuracy) is applied uniformly.

Table 7 shows that TanGCE achieves lower nonlinear leakage than AmbGCE consistently across all model families and sizes evaluated. The absolute gap narrows somewhat at larger scales but remains in the same direction, suggesting that the tangent projection's benefit is not an artifact of the particular hidden-state geometry of small models. Control accuracy changes (ΔCtrl) remain small and comparable between the two methods across all settings.

6.7 Second Vision Benchmark: FairFace

CelebA is a large-scale dataset of celebrity images, which may introduce distribution shifts specific to this demographic slice. To test cross-dataset robustness of our findings in the vision domain, we additionally evaluate on FairFace [Kärkkäinen and Joo, 2021], a balanced benchmark designed

Table 7: Multi-model generalization of TanGCE vs. AmbGCE at layer 8. NL: retrained nonlinear MLP probe accuracy on the target concept (lower = stronger erasure). ΔCtrl: downstream control-task accuracy change relative to clean representations (closer to zero = more surgical). λ^* selected per method from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL accuracy. TanGCE achieves lower NL leakage than AmbGCE consistently across all model families and sizes.

		Sycophancy		Gender (Bias in Bios)		Safety	
Model	Method	NL↓	ΔCtrl	NL↓	ΔCtrl	NL↓	ΔCtrl
Qwen2.5-0.5B	AmbGCE	.703	.000	.701	.000	.778	$-.040$
	TanGCE	**.602**	$-.015$	**.618**	$-.008$	**.663**	$+.039$
Qwen2.5-1.5B	AmbGCE	—	—	—	—	—	—
	TanGCE	—	—	—	—	—	—
Qwen2.5-3B	AmbGCE	—	—	—	—	—	—
	TanGCE	—	—	—	—	—	—
Gemma-2-2B	AmbGCE	—	—	—	—	—	—
	TanGCE	—	—	—	—	—	—
Mistral-7B	AmbGCE	—	—	—	—	—	—
	TanGCE	—	—	—	—	—	—

Table 8: CLIP ViT-B/32 concept erasure on FairFace (validation HPO for λ). FairFace provides diverse real-world face images with binary demographic labels (gender, binarized race groups, age). Each method's λ^* is selected from the same grid $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy. Controls: 5 least-correlated binary demographic attributes per concept. L/NL Leak: retrained linear / nonlinear probe accuracy (lower = better erasure). ΔCtrl: mean control accuracy change (closer to zero = more surgical). Embeddings: CLIP ViT-B/32 pooler_output (768 dimensions).

		Gender			Race (White)			Race (Black)			Age (Young)		
Method	λ^*	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	—	—	—	—	—	—	—	—	—	—	—	—	—
Random Per-Samp.	—	—	—	—	—	—	—	—	—	—	—	—	—
Linear Proj.	—	—	—	—	—	—	—	—	—	—	—	—	—
INLP	—	—	—	—	—	—	—	—	—	—	—	—	—
LEACE	—	—	—	—	—	—	—	—	—	—	—	—	—
AmbGCE	—	—	—	—	—	—	—	—	—	—	—	—	—
TanGCE	**—**	**—**	**—**	—	**—**	**—**	—	**—**	**—**	—	**—**	**—**	—

for diverse real-world face representation across gender, race, and age. We apply all seven erasure methods to CLIP ViT-B/32 `pooler_output` embeddings (768-dim) using the same validation HPO protocol as CelebA. For each of the four target concepts (Gender, Race White, Race Black, Age Young), we select 5 low-correlation binary demographic attributes as controls. Erasure strength λ^* is selected from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation nonlinear probe accuracy.

Table 8 shows that the pattern from CelebA replicates on FairFace: TanGCE achieves the lowest nonlinear leakage across all four demographic concepts while maintaining comparable control accuracy to AmbGCE. This confirms that the advantage of tangent-constrained erasure is not dataset-specific within the vision domain.

7 Discussion

The current results support a narrower central claim than the strongest possible version of the story. In every finished comparison, the tangent-based method removes more measured concept signal than its ambient counterpart. That pattern appears in the sycophancy sweeps, in the completed layer-8 safety sweep, and in the Bias in Bios diagnostic. What the present evidence does not yet show is the cleaner second half of the original thesis in which tangent projection also achieves lower displacement or uniformly better utility retention. In the completed runs, tangent reaches lower leakage by taking larger steps, and its utility metrics are either slightly worse or mixed.

The geometry diagnostics still matter because they explain why sycophancy is the harder case. Sycophancy shows local-vs-global angles near $80°$, while gender remains near $53°$, so a single global erasure direction is a much worse summary for the primary nonlinear concept. That is consistent with tangent-based methods improving removal in both concepts while leaving sycophancy as the clearly more nonlinear case, but the finished results currently frame tangent as a stronger remover rather than a dominantly better Pareto point.

Connection to low-rank fine-tuning. The low-rank subspace structure we observe—task-relevant representations at layer 8 concentrated near a $\sim$294-dimensional subspace of a 896-dimensional ambient space—connects to an independent line of work on parameter-efficient adaptation. Aghajanyan et al. [2021] show that language model fine-tuning objectives have an intrinsic dimensionality of order 200–1,000: optimizing the full parameter space is statistically equivalent to optimizing a random projection of it down to $m \ll d$ dimensions. Hu et al. [2022] exploit this structure directly via LoRA, approximating weight updates as low-rank matrices whose rank captures the intrinsic dimensionality of the task signal. Our finding is the activation-space analogue of the same phenomenon: just as parameter-space fine-tuning updates have low effective rank, the hidden-state directions that carry task-specific information span only a small linear subspace. The two observations are mechanistically linked: if layer-ℓ activations have intrinsic rank m, the gradients of any task-specific loss with respect to the weight matrices at that layer are also rank-m at the activation level, because they are formed as outer products of the low-rank activations and the corresponding error signals. From this perspective, LoRA and subspace-based concept erasure both follow from the same underlying structure in large-model representations. The key distinction is directionality: LoRA exploits low-rank structure to *add* task-specific behavior efficiently, while our approach exploits it to *remove* targeted concept information surgically.

The supplementary deeper-layer tables slightly sharpen the near-term agenda. In Tables 9 and 10, the same removal advantage continues at layers 16 and 23 for both sycophancy and safety, although the utility metrics differ by domain. We do not elevate those rows to headline claims because they are supplementary to the main layer-8 evaluation, but they strengthen the case for spending the next round of effort on matched-leakage comparisons and deeper-layer replications rather than on searching for a different primary mechanism.

Two implications follow. First, the paper should presently avoid claiming that off-manifold control is already proven to improve the full leakage-utility frontier; the matched-leakage analysis and broader replications are still needed for that claim. Second, the current evidence suggests that the next important question is not whether tangent projection changes the erasure direction at all, but under which layers and operating points that removal advantage can be converted into a cleaner utility tradeoff.

The matched layer-8 linear-family comparison clarifies one part of that tradeoff. LEACE is more surgical than tangent on this slice because it removes linearly decodable concept signal while taking a much smaller geometric step and preserving more downstream utility, especially on sycophancy where the answer-matching penalty is substantially smaller. That does not make LEACE a substitute for tangent on the main nonlinear claim: LEACE succeeds by solving the linear erasure problem cleanly, whereas tangent remains the stronger method when the target concept is evaluated with a fresh nonlinear probe. The current evidence therefore suggests a division of labor rather than a single winner: LEACE is the cleaner linear baseline, while tangent remains the more aggressive nonlinear remover.

8 Limitations

The main-text quantitative claims are deliberately limited to the core completed evaluation suite. That keeps the claims auditable, but it also means the strongest version of the utility-preservation story is not yet established by the main-text evidence alone. The deeper iterative sycophancy and safety comparisons are directionally consistent with the main narrative and are included in Appendix A, yet the paper's central claims still rely on the smaller completed suite.

A second limitation is measurement scope. The language-model evaluation now spans five models across three families, and the vision evaluation spans CelebA and FairFace with a shared CLIP encoder, but utility is still measured through task-specific proxies such as answer matching, helpfulness ranking,

profession classification, and control-attribute retention. These metrics are appropriate for surgicality, but they do not yet establish broad capability preservation across unrelated tasks.

Finally, the geometry diagnostics support low-dimensional structure, concept nonlinearity, and connectedness, but they do not by themselves prove that curved-manifold effects are the sole causal mechanism behind every leakage–utility tradeoff. The current paper therefore treats geometry as strong supporting evidence for tangent projection rather than as a complete causal proof.

9 Conclusion

We introduced TanGCE, the method presented in this paper: a manifold-aware concept erasure method that constrains erasure gradients to the local tangent space of the representation manifold. The central claim is not just that tangent projection removes more target information, but that it does so for the right geometric reason: when sentence representations occupy a much smaller natural manifold than the ambient hidden space, restricting updates to the tangent space lets the method use larger erasure strengths while keeping control-task changes small.

The comparison to IGBP and AmbGCE clarifies the role of the manifold constraint. IGBP is the closest prior nonlinear baseline; AmbGCE is the direct ablation with the same iterative loop but no tangent restriction; and TanGCE is the full method. The resulting λ sweep makes the practical benefit clear: ambient updates become unstable at large step sizes, whereas tangent-constrained updates continue to improve erasure precision.

Cross-modal experiments on CelebA and FairFace in CLIP ViT-B/32 representations confirm that TanGCE generalizes beyond language models, while the focused multi-model comparison shows that its advantage over AmbGCE persists across Qwen, Gemma, and Mistral backbones. Together, these results suggest that the tangent constraint is not tied to a single language-model family or a single face-image benchmark.

More broadly, the iterative nonlinear protocol follows a conservative lesson from prior erasure work: if a freshly trained classifier can still recover the concept, then the concept is still present. By repeatedly retraining the scorer and constraining each erasure step to the tangent space, TanGCE turns that lesson into a practical nonlinear erasure method.

Several directions remain open. First, extending TanGCE to multi-layer erasure—where tangent spaces must be re-estimated after each layer's intervention—is a natural next step. Second, while our experiments use PCA-based tangent estimation, learned manifold representations (e.g., variational autoencoders) could provide more accurate tangent spaces. Finally, the connection between local geometric structure and erasure effectiveness suggests that manifold-aware methods may benefit other representation editing tasks beyond concept removal.

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Table 9: Supplementary deeper-layer sycophancy comparison at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is answer-matching change relative to clean representations. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This table is included as supplementary evidence of directional consistency with the layer-8 main-text comparison in Table 2. Deeper-layer tangent runs continue to reduce leakage more than ambient baselines across this sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.063	0.213	-20.7pp	-0.5pp	2.88	0.62
16	0.153	0.400	+0.0pp	+0.0pp	3.79	0.27
23	0.138	0.361	-0.1pp	-0.2pp	44.50	3.11

Table 10: Supplementary deeper-layer safety comparison at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is pairwise helpfulness change. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This table is included as supplementary evidence of directional consistency with the layer-8 main-text comparison in Table 2. Deeper-layer tangent runs continue to reduce leakage more than ambient baselines across this sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.0725	0.2583	-0.052	-0.023	2.70	0.18
16	0.1292	0.3517	-0.092	+0.003	4.37	0.60
23	0.1483	0.3404	-0.062	-0.152	38.32	5.30

A Supplementary Deeper-Layer Comparisons

This appendix records broader deeper-layer comparisons that are directionally consistent with the main text but are not part of the core quantitative evidence used for the paper's principal claims. We include them to show how the same comparison pattern extends beyond the main layer-8 evaluation and to motivate future matched-leakage analyses.

A.1 Iterative Sycophancy (Supplementary Comparison)

A.2 Iterative Safety (Supplementary Comparison)

A.3 Local CelebA Visual Table-2-Style Comparison

To ensure a modality-matched benchmark, this appendix reports a vision-only CelebA run (CLIP ViT-B/32 embeddings) with the Table 2 method family and three least-correlated controls per target concept.

A.4 CelebA Per-Concept Detail (All 40 Attributes)

Tables 12 and 13 give the full per-concept breakdown behind the aggregate CelebA result in Table 5. The main-text Table 4 reports three representative concepts (Male, Smiling, Blond_Hair); the appendix tables cover all 40 binary CelebA attributes from the same CelebA all-40 evaluation setting under the shared validation-HPO protocol. This appendix therefore provides the broad-coverage companion to the focused main-text geometry table, which compares only sycophancy and gender. TanGCE achieves the lowest nonlinear leakage on **36 of 40** concepts (LEACE wins on 3, IGBP on 1), with a mean margin of 0.116 absolute NL accuracy over the best competing method per concept. Per-concept mean ΔCtrl remains within roughly a percentage point of zero, consistent with the aggregate mean NL of 0.707 and mean ΔCtrl of -0.7% reported in the main text.

Geometry Note. The broader geometry report continues to support three useful points: early and mid-layer representations are effectively low-dimensional, on-manifold directions carry much of the model's sensitivity in these perturbation tests, and interpolation remains consistent with a connected representation space. At the same time, the perturbation-sensitivity study is best read as

Table 11: Local Table-2-style CelebA visual comparison using CLIP ViT-B/32 embeddings. For each target concept we intervene once per method and evaluate (i) concept leakage via retrained linear/nonlinear probes (lower is better) and (ii) mean control-task accuracy change across 3 least-correlated control attributes (closer to zero is better). Targets and controls: Male $\rightarrow$ Blurry,Narrow_Eyes,Straight_Hair; Smiling $\rightarrow$ Straight_Hair,Bushy_Eyebrows,Gray_Hair; Blond_Hair $\rightarrow$ Straight_Hair,Narrow_Eyes,Blurry. Lambdas follow Table 2: all methods λ=0.5 except TanGCE λ=8.0.

	Male			Smiling			Blond_Hair		
Method	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl	L↓	NL↓	ΔCtrl
Random Global	.993	.994	-0.0pp	.891	.911	+0.0pp	.909	.933	-0.1pp
Random Per-Samp.	.993	.994	+0.0pp	.891	.907	+0.2pp	.907	.932	-0.1pp
Linear Proj.	.993	.992	+0.0pp	.900	.906	-0.3pp	.919	.931	-0.2pp
INLP	.994	.993	-0.0pp	.900	.903	+0.3pp	.918	.932	-0.0pp
LEACE	.994	.991	-0.0pp	.894	.894	+0.2pp	.909	.925	-0.1pp
AmbGCE	.990	.995	-0.2pp	.858	.884	-0.4pp	.893	.920	+0.2pp
TanGCE	.887	.953	-0.8pp	.649	.649	-0.6pp	.699	.747	-2.1pp

evidence of sensitivity concentration in a high-variance subspace rather than as a standalone proof of curved-manifold causality.

Table 12: Per-concept nonlinear probe leakage on CelebA (CLIP ViT-B/32), all 40 binary attributes. Each cell is retrained nonlinear probe test accuracy after erasure (lower = stronger erasure). λ^* is selected per method per concept from $\{0.5, 1.0, 2.0, 4.0, 8.0\}$ by lowest validation NL accuracy. Bold marks the lowest leakage per row. Abbrev.: RandG = Random Global, RandS = Random Per-Sample, LinProj = Linear Projection, AmbGCE = ambient gradient-based concept erasure, TanGCE = tangent gradient-based concept erasure.

Concept	RandG	RandS	LinProj	INLP	LEACE	IGBP	AmbGCE	TanGCE
5_o_Clock_Shadow	.92	.92	.91	.92	.87	.92	.91	**.77**
Arched_Eyebrows	.79	.79	.76	.76	.69	.77	.77	**.59**
Attractive	.73	.74	.69	.73	.63	**.53**	.73	.60
Bags_Under_Eyes	.76	.75	.74	.75	.70	.77	.75	**.69**
Bald	1.00	.99	.99	.99	.98	1.00	.99	**.97**
Bangs	.91	.91	.89	.89	.87	.92	.91	**.68**
Big_Lips	.80	.79	.74	.80	**.68**	.82	.81	.73
Big_Nose	.76	.76	.73	.74	.72	.73	.76	**.61**
Black_Hair	.85	.84	.82	.81	.79	.83	.82	**.46**
Blond_Hair	.92	.92	.94	.92	.92	.92	.92	**.72**
Blurry	.98	.98	.98	.96	.97	.98	.97	**.88**
Brown_Hair	.80	.80	.76	.75	.75	.80	.80	**.52**
Bushy_Eyebrows	.87	.88	.86	.85	.76	.87	.86	**.66**
Chubby	.94	.94	.95	.95	.92	.95	.94	**.81**
Double_Chin	.94	.94	.94	.93	.92	.94	.95	**.85**
Eyeglasses	.99	1.00	.96	.93	.96	1.00	.96	**.90**
Goatee	.96	.97	.97	.93	.96	.96	.95	**.85**
Gray_Hair	.97	.97	.97	.97	.97	.97	.98	**.89**
Heavy_Makeup	.89	.89	.89	.86	.85	.88	.88	**.70**
High_Cheekbones	.79	.79	.77	.75	.70	.57	.76	**.50**
Male	.99	.99	.99	.99	1.00	.98	1.00	**.85**
Mouth_Slightly_Open	.89	.89	.89	.71	.72	.66	.80	**.43**
Mustache	.94	.95	.95	.94	.92	.95	.93	**.77**
Narrow_Eyes	.88	.88	.80	.87	.80	.88	.88	**.78**
No_Beard	.94	.95	.94	.93	.92	.94	.95	**.77**
Oval_Face	.70	.71	.69	.68	**.67**	.69	.67	.72
Pale_Skin	.93	.93	.94	.95	.93	.92	.93	**.83**
Pointy_Nose	.75	.72	.71	.69	**.63**	.72	.75	.64
Receding_Hairline	.94	.94	.96	.96	.94	.94	.95	**.69**
Rosy_Cheeks	.93	.93	.93	.94	.91	.94	.93	**.79**
Sideburns	.95	.95	.94	.94	.95	.94	.94	**.82**
Smiling	.90	.89	.91	.74	.82	.59	.88	**.44**
Straight_Hair	.78	.78	.79	.77	.78	.78	.78	**.49**
Wavy_Hair	.78	.76	.77	.77	.78	.77	.78	**.52**
Wearing_Earrings	.90	.89	.86	.88	.85	.89	.88	**.61**
Wearing_Hat	1.00	1.00	.97	.97	.97	1.00	.97	**.94**
Wearing_Lipstick	.90	.90	.90	.89	.85	.88	.92	**.75**
Wearing_Necklace	.84	.85	.80	.80	.77	.85	.83	**.68**
Wearing_Necktie	.93	.93	.92	.92	.91	.92	.93	**.78**
Young	.86	.86	.85	.84	.83	.86	.87	**.60**
Mean	.88	.88	.87	.86	.84	.86	.87	**.71**

Table 13: Per-concept mean control accuracy change ΔCtrl (%) on CelebA, all 40 binary attributes. For each concept, ΔCtrl is averaged over the 5 least-correlated control attributes (closer to zero = more surgical). λ^* selection matches Table 12.

Concept	RandG	RandS	LinProj	INLP	LEACE	IGBP	AmbGCE	TanGCE
5_o_Clock_Shadow	+0.0	+0.1	-0.7	+0.1	+0.0	+0.1	-2.3	-0.2
Arched_Eyebrows	-1.3	-2.3	-1.2	-0.8	-1.1	-2.5	+0.6	-0.6
Attractive	-0.3	-1.0	-1.7	-0.8	-0.3	-0.9	+0.0	-0.8
Bags_Under_Eyes	-3.7	-1.7	-0.3	-7.7	-0.3	-0.3	-0.3	-0.1
Bald	+0.1	+0.0	-0.1	+0.1	-0.1	+1.1	+2.3	+0.2
Bangs	-3.7	-1.2	-0.5	-7.9	-0.1	-0.7	-0.3	-0.8
Big_Lips	+0.1	-0.0	-0.3	+0.0	-0.1	-0.5	+0.1	-0.1
Big_Nose	-0.1	+0.3	-3.5	-9.7	+0.0	-0.3	-1.8	+0.1
Black_Hair	-0.1	-0.0	+0.1	+0.2	+0.3	-0.3	-0.3	-0.7
Blond_Hair	-0.3	+0.1	-1.4	+0.7	-0.1	-0.4	-1.1	-1.6
Blurry	-1.3	+0.0	-0.1	-5.5	+0.1	-0.7	+0.7	-0.1
Brown_Hair	+0.1	-0.9	-0.3	-6.0	-0.7	+0.4	+0.1	-0.2
Bushy_Eyebrows	+0.0	+0.1	-0.1	-1.5	+0.2	-0.9	-0.3	-0.5
Chubby	-0.1	+0.3	-0.2	-1.4	-0.2	-0.2	+0.4	-2.0
Double_Chin	+0.1	+0.1	-0.4	-12.7	+0.3	+0.7	-0.7	-2.5
Eyeglasses	-0.1	+0.1	-0.0	-0.1	-0.2	-0.4	-0.2	+0.2
Goatee	+0.1	-0.1	+0.0	+0.2	+0.1	+0.5	-0.3	-0.4
Gray_Hair	-0.2	-0.3	-2.1	+0.4	-0.1	-1.1	-1.5	-1.5
Heavy_Makeup	-0.3	-0.1	-1.1	-1.7	-1.3	-1.3	-1.1	-0.9
High_Cheekbones	+0.1	+0.2	+0.1	-0.3	-0.1	-0.4	-0.0	-0.7
Male	-0.3	-0.1	-0.7	-0.2	-0.3	-2.2	-0.9	-1.1
Mouth_Slightly_Open	-0.3	-0.6	+0.1	+0.4	+0.2	+0.3	+0.1	-0.2
Mustache	+0.0	-0.1	+0.1	-10.3	-0.5	+1.0	+0.5	-1.5
Narrow_Eyes	-0.1	-0.2	-0.3	+0.2	-0.1	+0.3	+0.0	+0.0
No_Beard	-1.2	-0.2	-4.0	-6.5	-1.6	-0.1	-0.7	-0.5
Oval_Face	-2.0	+0.1	-0.1	-14.4	-0.6	-0.3	+0.2	-0.3
Pale_Skin	-1.2	-0.1	-0.1	-6.8	-0.9	-0.1	+0.7	-1.0
Pointy_Nose	-0.1	-0.2	+0.3	-1.1	+0.2	-0.7	-0.1	-0.1
Receding_Hairline	-0.1	-0.1	-0.1	-5.9	+0.0	-0.3	+0.1	-1.1
Rosy_Cheeks	-0.5	+0.1	-0.6	-0.1	-0.2	-0.1	-0.1	-1.3
Sideburns	-0.4	+0.0	-0.1	+0.1	+0.1	-0.6	-2.5	-0.1
Smiling	-0.3	-1.3	+0.1	+0.6	+0.4	+0.1	+0.1	-0.1
Straight_Hair	-0.1	-0.2	-0.1	-0.9	-0.1	-0.8	+0.5	-0.7
Wavy_Hair	+0.1	-0.7	-0.9	+0.2	+0.1	-0.4	-0.1	-0.7
Wearing_Earrings	+0.8	-0.3	-1.6	-0.3	-0.1	-1.3	-0.5	-0.3
Wearing_Hat	-0.1	+0.0	-0.1	-0.6	+0.0	+1.0	+0.1	-0.2
Wearing_Lipstick	+0.2	-0.1	-6.1	-21.5	-1.9	-1.2	-1.4	-0.9
Wearing_Necklace	+0.1	-1.7	-1.7	-9.3	+0.3	-1.8	+0.2	-1.2
Wearing_Necktie	+0.0	+0.1	+0.0	+0.2	-0.2	-0.7	-13.4	-1.3
Young	-0.0	-0.7	-1.7	-10.7	+0.1	-0.9	-2.2	-0.8
Mean	-0.4	-0.3	-0.8	-3.5	-0.2	-0.4	-0.6	-0.7

Table 14: Geometry evidence map organizing the appendix diagnostics into four research blocks. Each block addresses a distinct aspect of manifold structure: subspace sufficiency examines whether task-relevant computation concentrates in a low-dimensional linear subspace, perturbation sensitivity tests whether on-manifold directions matter more than orthogonal directions, representation connectivity evaluates whether the manifold remains connected across domains, and multi-layer accumulation tracks error compounding through the network. These diagnostics support the mechanism discussion and motivate the need for manifold-aware erasure methods but do not by themselves establish causality.

Experiment Block	Research Question	Primary Evidence
Subspace sufficiency	Subspace concentration	Top-1 agreement under intrinsic projection
Perturbation sensitivity	High-variance vs. orthogonal directions	KL and top-1 under on/off-subspace noise
Representation connectivity	Shared vs. domain manifolds	Cross-domain interpolation smoothness
Multi-layer accumulation	Error compounding across the network	Simultaneous intrinsic projection across layers

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